

So WSNs may have less computational power and less memory

4. Location information only

5. Communication is typically data-centric rather than address-centric, meaning routing may be aggregated/compressed/prioritised, even dropped, depending on the contents of the data

6. Communication in WSN takes place in very short packets to reduce overheads

7. Sensor networks have many-to-one traffic pattern, which may create hot-spot problems

Incorporating and taking care of the above features plays a tremendous role in designing protocols for efficient utilisation of the limited resources. Sensor networks operate under light load and suddenly become active in response to events like fire outbreak, earthquake or enemy movement.

WSN architecture

WSNs are a type of wireless networks that have three main components: large number of arbitrarily distributed sensor nodes, sink nodes (also known as base station or gateway) and software (see Fig. 1). Each node has a single omnidirectional antenna. Two sensors are neighbours if they are in transmission range of each other and can directly communicate with certain reliability. The gateway is located at the boundary of the monitored area, where it is reachable by at least some sensors and connects the micro-sensor network to the outside.

We refer to each micro-sensor as a data source or an event source and the gateway as a data sink or an event sink. The data is collected at the wireless sensor node, compressed and transmitted to the gateway directly or, if required, uses other wireless sensor nodes to forward data to the gateway. The transmitted data is then presented to the system by the gateway connection. The architecture of a micro-sensor contains four components: sensing circuitry, digital processing, power supply and radio transceiver.

Sensing circuitry. A wireless sensor node contains sensing and analogue-to-digital circuitry, which senses the surrounding environment and transforms it

into electrical signals. The sensed data may be about the behaviour of moving objects or static information like temperature, pressure and humidity. The position of the individual sensors cannot be predetermined. For example, the monitoring fields could be hostile or dangerous and may not be accessible in case of forest fire, chemical pollution, ongoing battle, etc.

A sensor network contains thousands of sensors, which are randomly and densely deployed (10 to 20 sensors per square metre). The sensors' range is not always a perfect circle, it could be an elliptical or some other shape. The sensing strength could vary with distance from the sensor. The development of sensor nodes equipped with Flash memory is giving a new direction in designing and deploying energy-efficient WSNs.

Sensor nodes can be imagined as small computers having limited computational power and limited memory. They consist of processing unit with sensors, designed to compute, calculate and distribute to communication device (transceiver or optical) power source).

The sink is a powerful workstation with plentiful resources whereas sensor nodes are low-power devices which have limited resources, memory space, computational capability and communication bandwidth. Desirable functions for sensor nodes include: ease of installation, self-identification, self-diagnosis, reliability, time awareness for coordination with other nodes, some software functions and digital signal processing (DSP), and standard control protocols and network interfaces. IEEE 1451 is the set of smart transducer interface standards developed by the Institute of Electrical and Electronics Engineers (IEEE) which meets these requirements.

As mentioned earlier, high degree of quality of service (QoS) or fidelity is important for sensor networks. QoS may be measured as percentage of the environment that is actually covered by active sensors. In a typical surveillance application, at least one sensor must remain active within one sub-region

of the network so that the intruder may be detected with high degree of probability. Since the main function of data sinks is acquiring the description of the environment rather than receiving all raw data collected by individual nodes, the throughput is less meaningful. This means, during operation, delay may be either much more or much less important.

Digital processing. The sensed data is fed to the processing unit, which usually has a small storage unit, for processing before transmission. This is referred to as 'in-network processing' during which redundant, useless and spurious data are deleted and observations from different sources are combined and aggregated. In-network data processing is a must to decrease the large volume of raw observations per sensor and to reduce the number of broadcasts. The data processing and storage capacity of the sensor node is limited.

Radio transceiver. A transceiver sends collected data via radio transmitter to sink node either directly or through intermediate sensor nodes. Sensor networks have short transmission range (up to 10 metres) and low data rate (several bytes). They may not have a globally unique ID.

Radio transceiver is the dominant power consumer. Digital radio transceivers for cellular communication backhaul and ground terminal transceivers for very small apertures terminals (VS-ATs) already employ m-band MMICs. Most VSATs operate in Ku band (12GHz to 18GHz) but, in the future, will be moving to Ka band (26GHz to 40GHz). However, most of the excitement lies in E-band (60GHz to 90GHz) which is meant for point-to-point WLANs and broadband Internet access. Point-to-point wireless can replace fiber-optic communication. Active antenna arrays and radar transmitters at W-Band, especially 94GHz, offer superior performance through clouds, fog and smoke.

Collision and congestion

If more than two nodes attempt to send data to the same destination at the same time, the destination node may fail to receive the data. This is called congestion.